Tackler and ball-carrier technique during moderate and severe injuries (≥8 days lost) compared with player-matched and team-matched injury-free controls in Elite Rugby Union

Vincent Meintjes,[1] Pip Forshaw,[1] Steve den Hollander ,[1,2] Lindsay Starling,[1] Michael Ian Lambert ,[1,2] Wayne Viljoen ,[1,2,3] Clint Readhead,[1,2,3] Sharief Hendricks [1,2,4]

[1]Division of Physiological Sciences, Department of Human Biology, Faculty of Health Sciences, University of Cape Town, Cape Town, South Africa
[2]Health, Physical Activity, Lifestyle, and Sport (HPALS) Research Centre, Department of Human Biology, Faculty of Health Sciences, University of Cape Town, Cape Town, South Africa
[3]Medical Department, South African Rugby Union, Cape Town, South Africa
[4]Carnegie Applied Rugby Research (CARR) centre, Institute for Sport, Physical Activity and Leisure, Leeds Beckett University, Leeds, UK

Correspondence to
Dr Sharief Hendricks, Department of Human Biology, Faculty of Health Sciences, University of Cape Town, Rondebosch 7725, Western Cape, South Africa; sharief.hendricks01@gmail.com

Accepted 30 June 2021

Check for updates

© Author(s) (or their employer(s)) 2021. No commercial re-use. See rights and permissions. Published by BMJ.

To cite: Meintjes V, Forshaw P, den Hollander S, *et al*. *Br J Sports Med* Epub ahead of print: [*please include* Day Month Year]. doi:10.1136/bjsports-2020-103759

ABSTRACT

Objective To analyse tackler and ball-carrier technical proficiency during moderate and severe contact injuries (≥8 days lost) in professional rugby union, and compare it with injury-free event-matched controls from the same player and from the same team.

Methods Technical proficiency for 74 (n=74) (moderate and severe; ≥8 days lost) tackler and ball-carrier injuries during The Currie Cup (2014–2018) and 623 matched non-injury events (253 own controls, 370 team controls) were examined through video analysis using a standardised list of technical criteria.

Results Mean technical proficiency score for injured tacklers during front-on tackles was 6.19/16 (arbitrary units (AU) 95% CI 4.89 to 7.48), which was significantly different to their own controls (8.90/16 AU, 95% CI 8.37 to 9.43, p<0.001, effect size (ES)=1.21, large) and team controls (9.93/16 AU, 95% CI 9.50 to 10.40, p<0.001, ES=1.71, large). Mean technical proficiency score for injured ball-carriers during front-on tackles was 5.60/14 AU (95% CI 4.65 to 6.55), which was significantly different to their own controls (8.08/14 AU, 95% CI 7.56 to 8.60, p<0.001, ES=1.16, moderate) and team controls (8.16/14 AU, 95% CI 7.75 to 8.57, p<0.001, ES=1.25, large).

Conclusion For the tackler and ball-carrier, for both front-on and side-on/behind tackles, overall technical proficiency scores were significantly lower for the injury-causing event, when compared with the player's own injury-free tackles and the team's injury-free tackles. Through analysing player and team controls, player technique deficiencies for the injured player and player technique deficiencies that expose all players to injury were highlighted, which may inform injury prevention strategies and policies, and assist coaches in optimising training to reduce tackle injury risk.

INTRODUCTION

The most frequently occurring contact event in rugby union is the tackle.[1 2] Tackle injuries have the highest injury incidence,[3] may cause the greatest number of days lost (severity)[4] and carry a high injury burden (injury incidence rate×mean days absent per injury).[5] Moderate and severe injuries in particular may have financial,[6] psychological[7] and team performance implications.[8] National and international governing bodies have the duty of care to protect their players and are therefore driven by improving player-welfare as well as achieving team success. In line with these objectives, reducing the risk of injury during tackle is a top priority.[9]

Studying 'what goes wrong' during injurious tackle events is key to developing effective injury prevention strategies.[10–12] Engaging in the tackle is a specialised technical skill, therefore studying players' technique during injurious events is a useful approach to understanding tackle injuries. One method to assess technique in the tackle is based on the diagnostic prescriptive approach,[13] where a player's technique is scored using a list of observable actions that represents the model form of the movement. The lists for the model form of the movement are derived from manuals and guidelines used by coaches to train safe and effective tackling and ball-carrying technique.[12 14–18] In other words, these are the technical criteria coaches are currently provided with to instruct players on how to safely and effectively perform the tackle. One point is awarded when a listed technique is performed and zero when the technique is not performed. The sum of these points represents the technical proficiency of the player. This method has been used to identify technical deficiencies in tackle injuries in youth rugby,[14 15] while at the professional level, technical deficiencies have been identified for head injuries or head impact tackles.[16–18]

A strength of all aforementioned studies is the inclusion of a control group. In the youth rugby studies, matched injury-free tackles of the same player were used as controls.[14 15] At the professional level, a random sample of tackles acted as controls.[16–18] Using the same player as a control helps identify technical deficiencies during the injurious event. However, interpreting these technical deficiencies may be confounded if the player had existing technical deficiencies. In other words, the player generally scores low for his/her technical proficiency, but scores even lower during an injury event—which places them at higher risk of injury compared with their teammates. This confounder can be addressed by also analysing the technical proficiency scores of non-injured team players as an additional control group.

Apart from the head injury and head impact studies,[16–18] tackle technical proficiency has not been analysed in professional rugby union. At the professional level, moderate and severe tackle

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.

injuries are of particular concern given its negative impact on player welfare and team success.[4–8] To reduce this negative impact, we need to further our understanding of the technical deficiencies during these injurious events, which will aid in the development of effective training interventions and/or policy changes. Therefore, the purpose of this study was to analyse tackler and ball-carrier technical proficiency during moderate and severe contact injuries in professional rugby union from 2014 to 2018, and compare it with injury-free event-matched controls from the same player and from the same team over the same period.

METHODS

Injury surveillance

The Currie Cup Premier division is South Africa's annual premier domestic senior male professional level rugby union competition. Injury surveillance was conducted during The Currie Cup between 2014 and 2018 as part of the SA Rugby Injury and Illness Surveillance and Prevention Project (SARIISPP).[19] Each team participating in the tournament had an appointed team doctor or medical support personnel, who captured all the relevant match injury data on a bespoke application using an electronic device. The application provided the medical personnel with a standardised injury surveillance data capture format prepared by BokSmart,[19] which is based on the International Consensus statement for injury surveillance in rugby union.[20] The injury definition for the current study was therefore aligned with the International Consensus statement,[20] and was only concerned with time loss injuries recorded as 'moderate' and 'severe' severity (≥8 days' time loss). All captured injury data were stored in the South African Rugby Union (SARU) Injury Database (HREC 438/2011). All players competing in the tournament provided consent to have their injury data collected, stored and analysed. For this study, all moderate and severe tackle and ruck injuries (≥8 days' time loss) were retrieved from the database.

Injury incidence

A total of 443 players were injured (579 injuries) between 2014 and 2018 (86 players in 2014, 93 in 2015, 105 in 2016, 93 in 2017 and 66 in 2018). The resultant injury rates (per 1000 player hours) were 91 (95% CI 75 to 107), 66 (95% CI 54 to 78), 91 (95% CI 76 to 106), 81 (95% CI 67 to 95) and 82 (95% CI 64 to 100) for 2014, 2015, 2016, 2017 and 2018, respectively.

Video analysis

Using the SARIISPP injury surveillance data described above (specifically, the team, playing position and match quarter data), video footage for each injury was sourced for further analysis from the University Video Analysis Database (HREC R023/2017). For each injury, we set out to identify two control groups—non-injury tackle events by the injured player (own controls) and matched non-injury tackle events by a teammate (team controls). While a 1:2:2 ratio (for each injury, two controls for each control group) would be considered sufficient,[21] we wanted both control groups to be highly representative. Accordingly, matched injury-free tackle events within the same or most recent match prior to the injury were identified. A ratio of 1:5:5 was set as the goal (10 control events in total), however, identifying controls was dependent on how frequently the matched injury-free tackle events occurred within the same or most recent match. Both own control and team control tackle events were matched by tackle type, tackle direction and role (ball-carrier or tackler). All control tackles (for both control groups) were identified retrospectively from the point of injury until the 1:5:5 ratio was reached. When a 1:5:5 ratio was not possible, all matched injury-free own controls and team controls within the same or most recent match prior to the injury were analysed.

A tackle was defined as 'when one or more tacklers (player or players making the tackle) attempt to stop or impede the ball-carrier (player carrying the ball), irrespective of whether or not the ball-carrier goes to ground'.[12 22 23] Shoulder and arm tackles are the most frequently occurring tackles during a match,[22] and to date, only technical criteria for these types of tackles have been validated.[12 14–18 24] As such, only shoulder and arm tackles for front-on and side/behind tackles were analysed. Tackles were analysed based on the injured players role in the tackle, that is, ball-carrier or primary tackler.

Matching injury to video

Of the 579 injuries, 288 had a severity of ≥8 days lost (moderate and severe injuries); of which 74 (31 tackler, 43 ball-carrier) were available for video analysis. A total of 623 matched non-injury events were identified, which equated to 256 tackler (101 own controls, 155 team controls) and 367 ball-carrier (152 own controls, 215 team controls) (figure 1).

Ruck technique has only been analysed in one study on youth concussions.[15] Originally, analysing ruck technique was part of the original objectives of this study. However, only three moderate and severe injuries (n=3) met the criteria for technique analyses. As a result, it was decided to remove the ruck technique analyses from the study.

All video footage was analysed using Sports Code elite V.11.3.0 (Hudl, Lincoln, Nebraska, USA) using an Apple iMac (Apple, Cupertino, California, USA). The analysis software allowed for the recording and saving of each coded instance into a database. During the analysis, the analyst could pause, rewind and watch

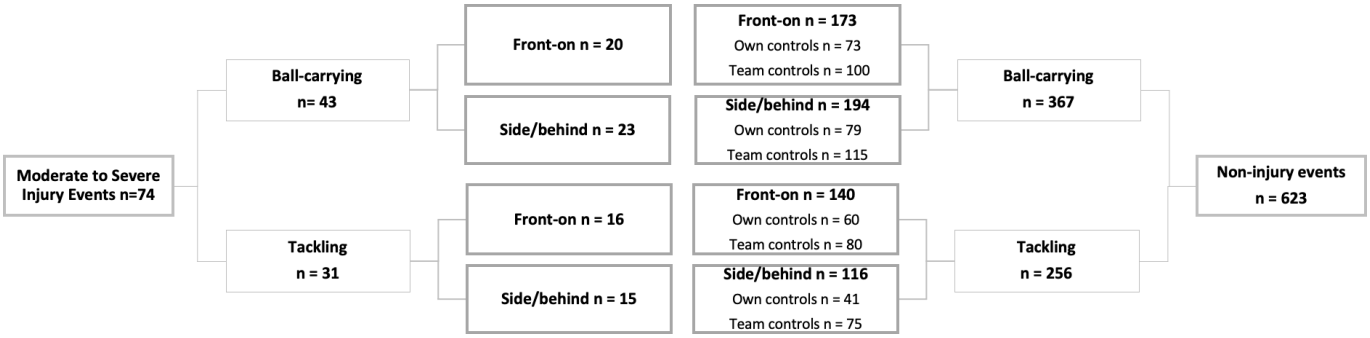

Figure 1 Breakdown of the number of selected tackle injury events, and player-matched (own controls) and team-matched (team controls) non-injury events. Tackle events are categorised according to the player's role and the direction of the tackle.

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.

the footage in slow motion. A single analyst assessed the technical proficiency for all ball-carrier and tackler events.

Contact proficiency for the tackler and ball-carrier were assessed using a standardised list of technical criteria, which are based on a list of observable actions that represents the model form of the movement and are used to coach techniques for tackling and carrying the ball into contact.[12 14–18] The technical criteria were agreed on by a panel of coaches, trainers, sport clinicians and sport and exercise scientists. Recently, the technical criteria also showed strong agreement among the Rugby Video Analysis Consensus Group.[12] Tackler and ball-carrier criteria were split into three phases—precontact, contact and postcontact. During the technique analysis, a player was awarded either one point or zero depending on whether a particular action was performed or not. The sum of technical criteria met by the player was subsequently used to represent the technical proficiency of the player and reported in arbitrary units (AU). The technical criteria for a front-on tackler (maximum total score 16) and side-on/behind tackler (maximum total score 11) are shown in table 1; and the technical criteria for a front-on ball-carrier (maximum total score 14) and side-on/behind ball-carrier (maximum total score 12) are shown in table 2.

Reliability

Ten players (four tacklers and six ball-carriers) were randomly selected for intra-reliability testing.[12 25 26] This equated to 44 tackler events and 66 ball-carrier events (n=110 tackle events). Intra-reliability- and inter-reliability were tested using the a two-way random-effects intraclass correlation coefficient (ICC)[27] and the typical error of measurement (TEM), which we report as a percentage (TEM %).[28] Intra-reliability for tacklers was ICC=0.97, TEM %=11.4%, and ICC=0.99, TEM=9.8% for ball-carriers. To test the validity of the technical criteria, an external coder followed the same procedure on a sample of tackles (18 tackler events and 19 ball-carrier events) to determine inter-reliability. For tacklers, the ICC and TEM % for inter-reliability were 0.92% and 12.9%, respectively. For ball-carriers, the ICC and TEM % for inter-reliability were 0.78% and 17.7%, respectively. These intra-reliability and inter-reliability values are comparable to previous studies scoring technical proficiency for tackle injury events.[14 15]

Statistical analyses

Data were checked for normality using the Shapiro-Wilk normality test. Means with 95% CIs for each technical criterion, each phase of the event and overall technique scores for tackle and ball-carrier are reported, including total percentage scores. One-way analysis of variance was used to compare the three groups (injury, own controls and team controls) with separate analyses computed for tacklers and ball-carriers, and for front-on and side-on tackles. Post hoc analyses were conducted using the Tukey's multiple comparison test. A two-tailed p value was used for all tests, with the a priori alpha level of significance set at $p<0.05$. The mean difference and 95% CI of the difference, as well as overlapping 95% CI were also examined. Cohen's effect sizes (ES) were calculated to determine the magnitude of the differences between groups with an ES of <0.19, 0.20–0.59, 0.60–1.19 and >1.2 considered trivial, small, moderate and large, respectively.[29] Only differences that were statistically significant and had a moderate or large ES (ES >0.60) were considered. Injury data are reported per 1000 player hours with 95% CI. Statistical analyses and graphs were computed using STATA V.13 (StataCorp, College Station, Texas, USA) and Prism V.8 (GraphPad, San Diego, California, USA).

RESULTS

Tackler proficiency

The mean technical proficiency score for injured tacklers during front-on tackles was 6.19/16 AU (95% CI 4.89 to 7.48) (figure 2). This was significantly different to their own controls (8.90/16 AU, 95% CI 8.37 to 9.43, $p<0.001$, ES=1.21, large) and team controls (9.93/16 AU, 95% CI 9.50 to 10.40, $p<0.001$, ES=1.72, large). Players' own control technical proficiency scores were also significantly different to their team's technical proficiency scores ($p<0.01$, ES=0.52, small). Differences between groups for each front-on tackler technical criterion are shown in table 1.

The mean technical proficiency score for injured tacklers during side-on/behind tackles was 4.80/11 AU (95% CI 3.91 to 5.69) (figure 2). This was significantly different to their own controls (6.44/11 AU, 95% CI 5.89 to 6.99, $p<0.01$, ES=0.98, moderate) and team controls (7.23/11 AU, 95% CI 6.85 to 7.60, $p<0.001$, ES=1.50, large). Players' own control technical proficiency scores were also significantly different to their team's technical proficiency scores ($p<0.05$, ES=0.47, small). Differences between groups for each side-on/behind tackler technical criterion are shown in table 1.

Ball-carrier proficiency

The mean technical proficiency score for injured ball-carriers during front-on tackles was 5.60/14 AU (95% CI 4.65 to 6.55) (figure 2). This was significantly different to their own controls (8.08/14 AU, 95% CI 7.56 to 8.60, $p<0.001$, ES=1.16, moderate) and team controls (8.16/14 AU, 95% CI 7.75 to 8.57, $p<0.001$, ES=1.25, large).

For side/behind tackled ball-carriers, total proficiency scores for the injury group (5.17/12 AU, 95% CI 4.23 to 6.12) were significantly different to their own controls (7.39/12 AU, 95% CI 7.00 to 7.79, $p<0.001$, ES=1.11, moderate) and team controls (7.75/12 AU, 95% CI 7.48 to 8.02, $p<0.001$, ES=1.38 large) (figure 2).

Ball-carriers' own control technical proficiency scores were not significantly different to their team's technical proficiency scores for front-on ($p>0.05$, ES=0.04, trivial) and side/behind tackles ($p>0.05$, ES=0.22, small). Differences between groups for each front-on and side-on/behind technical ball-carrier criterion are shown in table 2.

DISCUSSION

This study is novel on three fronts: (i) it is the first study to analyse tackler and ball-carrier technique proficiency in professional rugby union matches, outside of head injuries; (ii) it focuses on moderate and severe tackler and ball-carrier injuries collected over 4 years and (iii) it includes two 'control' groups—the player's own injury-free tackles and ball-carries, and injury-free tackles and ball-carries from their teammates. The findings of this study may potentially lead to more effective training interventions and/or policy changes, and as a result, protect players from injury and improve their welfare (figure 3).

For the tackler and ball-carrier, for both front-on and side-on/behind tackles, overall technical proficiency scores were significantly lower for the injury-causing event, both when compared with the player's own injury-free tackles, and the team's injury-free tackles.

Given the severity of these injuries, these findings reinforce the argument that poor contact technique is a risk factor for injury, that is, proper tackler and ball-carrier technique will reduce players' risk of injury when contesting the tackle.[30 31] The best approach to develop and improve contact technique in the tackle contest is through adequate skills training.[32 33] However, a study on 1.5

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.

Table 1 Tackler front-on and side/behind proficiency scores for injuries (I), own controls (OC) and team controls (TC)

	I		OC		TC		I versus OC		I versus TC		OC versus TC	
	Mean	95% CI	Mean	95% CI	Mean	95% CI	Effect size	Interpretation	Effect size	Interpretation	Effect size	Interpretation
Tackler front-on	(n=16)		(n=60)		(n=80)							
Precontact	3.75	3.15 to 4.35	5.23	4.95 to 5.52	5.38	5.15 to 5.60	1.33	Large***	1.53	Large***	0.14	Trivial
Identify ball-carrier onto shoulder	0.81	0.60 to 1.00	0.95	0.89 to 1.00	0.98	0.94 to 1.00	0.43	Small	0.56	Small*	0.16	Trivial
Body position: upright to low	**0.56**	**0.29 to 0.84**	**0.50**	**0.37 to 0.63**	**0.86**	**0.79 to 0.94**	**0.12**	**Trivial**	**0.69**	**Moderate***	**0.83**	**Moderate*****
Keep back straight, centre of gravity ahead of base support	**0.25**	**0.01 to 0.49**	**0.75**	**0.64 to 0.86**	**0.56**	**0.45 to 0.67**	**1.13**	**Moderate*****	**0.65**	**Moderate***	**0.41**	**Small**
Alignment square to ball-carrier	0.44	0.16 to 0.71	0.68	0.56 to 0.80	0.89	0.82 to 0.96	0.49	Small	1.06	Moderate***	0.52	Small*
Assume 'boxer stance' (elbows low and close, hands up)	0.13	0.00 to 0.31	0.33	0.21 to 0.46	0.29	0.19 to 0.39	0.48	Small	0.40	Small	0.09	Trivial
Head up and face forward	0.50	0.22 to 0.78	0.85	0.76 to 0.94	0.75	0.65 to 0.85	0.79	Moderate**	0.52	Small	0.25	Small
Shortening steps	0.06	0.00 to 0.20	0.17	0.07 to 0.26	0.05	0.00 to 0.10	0.34	Small	0.04	Trivial	0.39	Small
Approach from front/oblique	1.00	1.00 to 1.00	1.00	1.00 to 1.00	1.00	1.00 to 1.00	–	–	–	–	–	–
Contact	1.81	1.16 to 2.46	1.98	1.71 to 2.25	2.48	2.27 to 2.68	0.15	Trivial	0.62	Moderate	0.51	Small*
Explosiveness (rapid movement) on contact	0.38	0.11 to 0.64	0.35	0.23 to 0.47	0.25	0.15 to 0.35	0.06	Trivial	0.28	Small	0.22	Small
Contact with shoulder	**0.56**	**0.29 to 0.84**	**0.55**	**0.42 to 0.68**	**0.91**	**0.85 to 0.98**	**0.02**	**Trivial**	**0.85**	**Moderate****	**0.88**	**Moderate*****
Contact in centre of gravity	0.31	0.06 to 0.57	0.23	0.12 to 0.34	0.43	0.31 to 0.54	0.18	Trivial	0.25	Small	0.43	Small
Head placement on correct side of ball-carrier	**0.56**	**0.29 to 0.84**	**0.85**	**0.76 to 0.94**	**0.89**	**0.82 to 0.96**	**0.66**	**Moderate***	**0.77**	**Moderate****	**0.12**	**Trivial**
Postcontact	0.63	0.08 to 1.17	1.68	1.43 to 1.93	2.08	1.87 to 2.28	1.05	Moderate***	1.48	Large***	0.42	Small
Shoulder drive on first contact	0.25	0.01 to 0.49	0.32	0.20 to 0.44	0.45	0.34 to 0.56	0.15	Trivial	0.42	Small	0.27	Small
Leg drive on contact	0.06	0.00 to 0.20	0.15	0.06 to 0.24	0.16	0.08 to 0.25	0.29	Small	0.32	Small	0.03	Trivial
Arm usage—rapidly extends arms forward, wrap and pull (hit and stick)	**0.25**	**0.01 to 0.49**	**0.60**	**0.47 to 0.73**	**0.76**	**0.67 to 0.86**	**0.74**	**Moderate***	**1.17**	**Moderate*****	**0.35**	**Small**
Release ball-carrier and compete for possession (jackaling)	**0.06**	**0.00 to 0.20**	**0.62**	**0.49 to 0.74**	**0.70**	**0.60 to 0.80**	**1.44**	**Large*****	**1.73**	**Large*****	**0.17**	**Trivial**
Total/16	6.19	4.89 to 7.48	8.90	8.37 to 9.43	9.93	9.50 to 10.40	1.21	Large***	1.72	Large***	0.52	Small**
%	39		56		62							
Tackler side/behind	(n=15)		(n=41)		(n=75)							
Precontact	2.67	2.21 to 3.12	3.17	2.85 to 3.49	3.43	3.21 to 3.65	0.54	Small	0.85	Moderate*	0.26	Small
Identify ball-carrier onto shoulder	0.87	0.67 to 1.00	0.88	0.77 to 0.98	0.97	0.94 to 1.00	0.03	Trivial	0.36	Small	0.35	Small
Body position: upright to low	0.67	0.40 to 0.94	0.49	0.33 to 0.65	0.69	0.59 to 0.80	0.36	Small	0.04	Trivial	0.41	Small
Keep back straight, cog ahead of base support	0.53	0.25 to 0.82	0.78	0.65 to 0.91	0.65	0.54 to 0.76	0.53	Small	0.24	Small	0.29	Small
Head up and face forward	**0.53**	**0.25 to 0.82**	**0.85**	**0.74 to 0.97**	**0.84**	**0.76 to 0.92**	**0.72**	**Moderate***	**0.69**	**Moderate***	**0.03**	**Trivial**
Shortening steps	0.07	0.00 to 0.21	0.17	0.05 to 0.29	0.27	0.16 to 0.37	0.31	Small	0.55	Small	0.24	Small
Contact	1.67	1.32 to 2.01	1.61	1.38 to 1.84	1.99	1.81 to 2.17	0.09	Trivial	0.46	Small	0.50	Small*
Explosiveness (rapid movement on contact)	0.47	0.18 to 0.75	0.34	0.19 to 0.49	0.57	0.46 to 0.69	0.26	Small	0.20	Small	0.47	Small
Contact in centre of gravity	0.40	0.12 to 0.68	0.34	0.19 to 0.49	0.45	0.34 to 0.57	0.12	Trivial	0.10	Trivial	0.22	Small
Head placement on correct side/behind ball-carrier	0.80	0.57 to 1.00	0.93	0.84 to 1.00	0.96	0.91 to 1.00	0.37	Small	0.49	Small	0.13	Trivial
Postcontact	0.47	0.00 to 0.93	1.66	1.42 to 1.90	1.81	1.64 to 1.99	1.49	Large***	1.67	Large***	0.20	Small
Shoulder driven on first contact	0.13	0.00 to 0.33	0.22	0.09 to 0.35	0.39	0.27 to 0.50	0.23	Small	0.61	Moderate	0.37	Small
Arm usage—rapidly extends arms forward, wrap and pull (hit and stick)	**0.27**	**0.01 to 0.52**	**0.76**	**0.62 to 0.89**	**0.91**	**0.84 to 0.97**	**1.10**	**Moderate*****	**1.66**	**Large*****	**0.40**	**Small**
Release ball-carrier and compete for possession (jackaling)	**0.07**	**0.00 to 0.21**	**0.68**	**0.53 to 0.83**	**0.52**	**0.40 to 0.64**	**1.61**	**Large*****	**1.13**	**Moderate****	**0.33**	**Small**
Total /11	4.80	3.91 to 5.69	6.44	5.89 to 6.99	7.23	6.85 to 7.60	0.98	Moderate**	1.50	Large***	0.47	Small*
%	44		59		66							

Data are reported as means with 95% CI. Total percentage scores, effect sizes with interpretations and significant levels are also provided.
*P≤0.05; **p≤0.01; ***p≤0.001.

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.

Table 2 Ball-carrier front-on and side/behind proficiency scores for injuries (I), own controls (OC) and team controls (TC)

	I		OC		TC		I versus OC		I versus TC		OC versus TC	
	Mean	95% CI	Mean	95% CI	Mean	95% CI	Effect size	Interpretation	Effect size	Interpretation	Effect size	Interpretation
Ball-carrier front on	(n=20)		(n=73)		(n=100)							
Precontact	2.85	2.39 to 3.31	3.78	3.49 to 4.07	4.15	3.97 to 4.33	0.83	Moderate***	1.38	Large***	0.34	Small
Focus on tackler	0.85	0.68 to 1.00	0.88	0.80 to 0.95	0.97	0.94 to 1.00	0.09	Trivial	0.42	Small	0.34	Small
Shift ball away from contact to correct arm	**0.30**	**0.08 to 0.52**	**0.64**	**0.53 to 0.76**	**0.66**	**0.57 to 0.75**	**0.71**	**Moderate***	**0.76**	**Moderate****	**0.04**	**Trivial**
Body position: upright to low	0.50	0.26 to 0.74	0.58	0.46 to 0.69	0.51	0.41 to 0.61	0.16	Trivial	0.02	Trivial	0.14	Trivial
Keep back straight and centre of gravity ahead of base of support	0.50	0.26 to 0.74	0.66	0.55 to 0.77	0.83	0.76 to 0.90	0.32	Small	0.73	Moderate**	0.39	Small*
Head up and face forward	0.65	0.42 to 0.88	0.73	0.62 to 0.83	0.89	0.83 to 0.95	0.17	Trivial	0.58	Small*	0.41	Small*
Shuffle or evasive manoeuvre	0.05	0.00 to 0.15	0.30	0.19 to 0.41	0.29	0.20 to 0.38	0.69	Moderate	0.67	Moderate	0.02	Trivial
Contact	1.65	1.06 to 2.24	2.68	2.42 to 2.95	2.40	2.18 to 2.62	0.86	Moderate***	0.62	Moderate*	0.25	Small
Fending into contact	0.20	0.01 to 0.39	0.40	0.28 to 0.51	0.32	0.23 to 0.41	0.44	Small	0.27	Small	0.17	Trivial
Side-on into contact	0.15	0.00 to 0.32	0.45	0.34 to 0.57	0.33	0.24 to 0.42	0.68	Moderate*	0.45	Small	0.25	Small
Explosiveness (rapid movement) on contact	0.65	0.42 to 0.88	0.74	0.64 to 0.84	0.64	0.54 to 0.74	0.19	Trivial	0.02	Trivial	0.22	Small
Body position: from low up into contact	0.15	0.00 to 0.32	0.30	0.19 to 0.41	0.29	0.20 to 0.38	0.36	Small	0.34	Small	0.02	Trivial
Ball in correct arm and protected	**0.50**	**0.26 to 0.74**	**0.79**	**0.70 to 0.89**	**0.82**	**0.74 to 0.90**	**0.63**	**Moderate***	**0.70**	**Moderate****	**0.08**	**Trivial**
Postcontact	1.10	0.73 to 1.47	1.62	1.47 to 1.76	1.61	1.45 to 1.77	0.74	Moderate*	0.65	Moderate*	0.01	Trivial
Leg drive on contact	0.20	0.01 to 0.39	0.45	0.34 to 0.57	0.34	0.25 to 0.43	0.55	Small	0.32	Small	0.23	Small
Arm usage-use arm and shoulder to push tackler	0.20	0.01 to 0.39	0.26	0.16 to 0.36	0.32	0.23 to 0.41	0.14	Trivial	0.27	Small	0.13	Trivial
Go to ground and present ball/break tackle/offload	0.70	0.48 to 0.92	0.90	0.83 to 0.97	0.95	0.91 to 0.99	0.51	Small*	0.68	Moderate***	0.19	Trivial
Total/14	5.60	4.65 to 6.55	8.08	7.56 to 8.60	8.16	7.75 to 8.57	1.16	Moderate***	1.25	Large***	0.04	Trivial
%	40		58		58							
Ball-carrier side/behind	(n=23)		(n=79)		(n=115)							
Precontact	3.09	2.44 to 3.74	4.22	3.99 to 4.44	4.31	4.16 to 4.47	0.88	Moderate***	1.00	Moderate***	0.10	Trivial
Aware of tackler/attunement	**0.70**	**0.49 to 0.90**	**1.00**	**1.00 to 1.00**	**0.96**	**0.92 to 0.99**	**0.90**	**Moderate*****	**0.72**	**Moderate*****	**0.28**	**Small**
Shifting the ball away from contact to correct arm	0.52	0.30 to 0.74	0.59	0.48 to 0.71	0.73	0.65 to 0.81	0.14	Trivial	0.44	Small	0.30	Small
Body position: upright to low	0.22	0.04 to 0.40	0.42	0.31 to 0.53	0.40	0.31 to 0.49	0.43	Small	0.39	Small	0.04	Trivial
Keep back straight and centre of gravity ahead of base of support	0.65	0.44 to 0.86	0.82	0.74 to 0.91	0.89	0.83 to 0.95	0.39	Small	0.58	Small*	0.20	Small
Head up and face forward	0.74	0.54 to 0.93	0.82	0.74 to 0.91	0.91	0.86 to 0.97	0.19	Trivial	0.45	Small	0.27	Small
Shuffle or evasive manoeuvre	0.26	0.07 to 0.46	0.56	0.44 to 0.67	0.43	0.33 to 0.52	0.63	Moderate*	0.36	Small	0.26	Small
Contact	1.13	0.80 to 1.46	1.61	1.43 to 1.78	1.78	1.65 to 1.91	0.63	Moderate*	0.89	Moderate***	0.23	Small
Fending into contact	0.30	0.10 to 0.51	0.32	0.21 to 0.42	0.35	0.26 to 0.44	0.04	Trivial	0.11	Trivial	0.06	Trivial
Explosiveness (rapid movement) on contact	**0.17**	**0.01 to 0.34**	**0.46**	**0.34 to 0.57**	**0.61**	**0.52 to 0.70**	**0.65**	**Moderate***	**1.00**	**Moderate*****	**0.30**	**Small**
Ball in correct arm and protected	0.65	0.44 to 0.86	0.84	0.75 to 0.92	0.83	0.76 to 0.90	0.44	Small	0.41	Small	0.03	Trivial
Postcontact	0.96	0.68 to 1.23	1.57	1.40 to 1.74	1.65	1.52 to 1.78	0.87	Moderate***	1.04	Moderate***	0.11	Trivial
Leg drive on contact	0.26	0.07 to 0.46	0.33	0.22 to 0.44	0.45	0.36 to 0.54	0.15	Trivial	0.40	Small	0.25	Small
Arm usage—use arm and shoulder to push tackler	0.09	0.00 to 0.21	0.30	0.20 to 0.41	0.25	0.17 to 0.33	0.54	Small	0.43	Small	0.11	Trivial
Go to ground and present ball/break tackle/offload	**0.61**	**0.39 to 0.82**	**0.94**	**0.88 to 0.99**	**0.95**	**0.91 to 0.99**	**0.84**	**Moderate*****	**0.88**	**Moderate*****	**0.04**	**Trivial**
Total/12	5.17	4.23 to 6.12	7.39	7.00 to 7.79	7.75	7.48 to 8.02	1.11	Moderate***	1.38	Large***	0.22	Small
%	43		62		65							

Data are reported as means with 95% CI. Total percentage scores, effect sizes with interpretations and significant levels are also provided.
*P≤0.05; **p≤0.01; ***p≤0.001.

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.

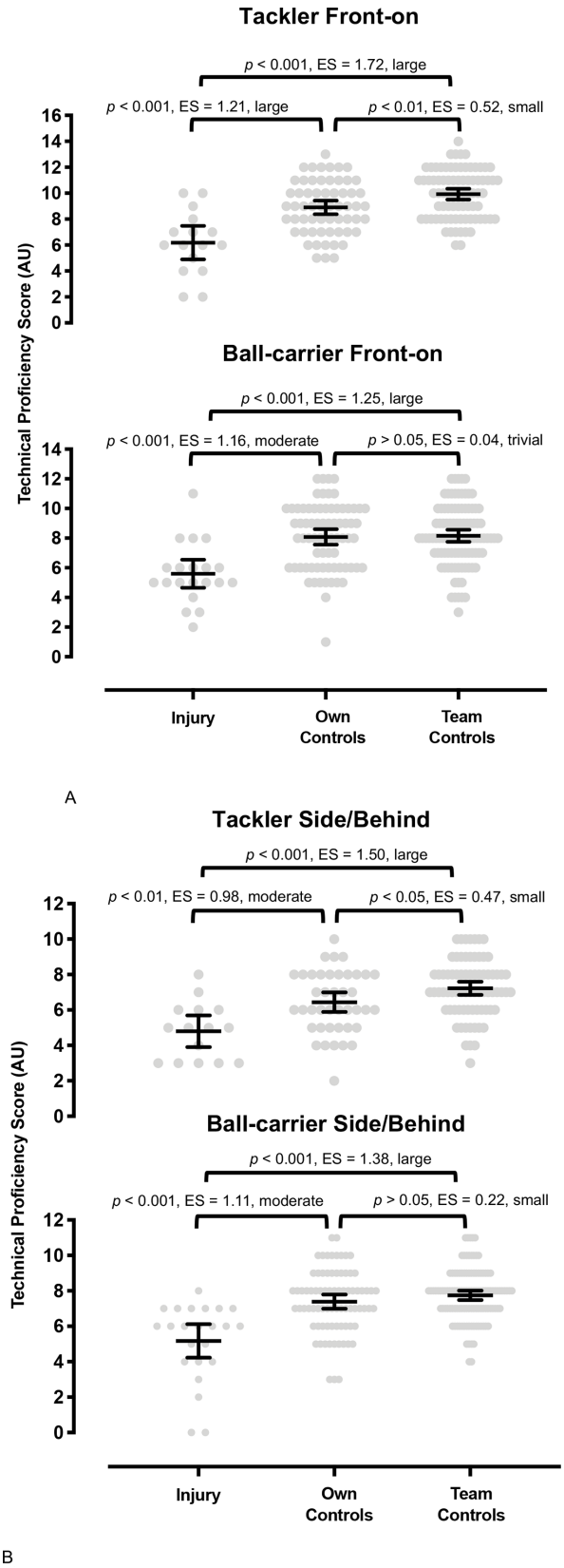

Figure 2 (A) Technical proficiency scores for front-on ball-carriers and tacklers during the injury event, and player-matched (own controls) and team-matched (team controls) non-injury events. Solid lines represent means with 95% CIs. (B) Technical proficiency scores for side/behind ball-carriers and tacklers during the injury event, and player-matched (own controls) and team-matched (team controls) non-injury events. Solid lines represent means with 95% CIs. ES, effect size.

million hours of training exposure over 11 seasons in elite rugby union suggests players are only spending a small percentage of their total weekly training on contact training (with the majority of the time spent on gym conditioning, non-gym-based conditioning and non-contact training)[34]—and we can only assume part of that small percentage is focused on technique. Also, there seems to be a mismatch between tackle contact training preparation and tackle contact match demands in elite rugby union.[35] Considering these training studies and in light of our technique findings, rethinking how we train the tackle will strengthen player welfare and injury prevention efforts.

Techniques which expose the ball-carrier and tackler to injury when misperformed

Having two control groups also helps determine which techniques expose all players to injury, and have both tackle law and training implications. These techniques are identified when the injured group scored significantly lower (with a moderate to large ES) than the player's own control and the team control—but the same technique was not different between the control groups. For the tackler, the techniques which were lacking were—keeping the back straight (front-on) while placing the centre of gravity ahead of base support (front-on), placing the head on correct side of ball-carrier (front-on tackle), facing the head up and forward (side-on/behind), using the arms after contact (rapidly extending the arms forward, with a wrap and pull, ie, hit and stick) (both types of tackles) and releasing the ball-carrier and competing for possession (both types of tackles). These specific tackler techniques also scored significantly lower than non-injury events in youth rugby and injurious head tackles.[14 16 18] For instance, in a sample of 74 tackles that resulted in a head injury assessment for the tackler, Tierney *et al*[16] found tackler's keeping a back straight with the centre of gravity ahead of base support, placing the head on correct side of ball-carrier, facing the head up and forward and using the arms after contact to have a lower propensity for a head injury assessment. The agreement between which specific tackler techniques are lacking during injurious events in this study (for moderate and severe injuries) and head injury studies,[16 18] provides strong evidence for the importance of these techniques in reducing the risk of serious injuries. As such, these tackler techniques need to be emphasised during tackle training, and when providing player and coach injury prevention education.

Ball-carrier techniques that scored significantly lower than non-injury events were—awareness of the tackler (side-on/behind), shifting the ball away from contact to the correct arm (front-on), explosiveness on contact (side-on/behind), protecting the ball in the correct arm (front-on) and going to ground and presenting ball/break tackle/offload (side-on/behind). Burger *et al*[14] also found the ball-carrier to be less proficient during injurious side-on tackles in high-level under 18 players. The study by Burger *et al*[14] analysed tackle injuries over 3 years (2011–2013) in the under 18 Craven Week, a tournament for South Africa's top schoolboy players. This tournament acts as one of the main competitions for young players to be selected into the senior provincial teams, and subsequently compete in The Currie Cup. Considering this relationship between the two tournaments, and the similarities in ball-carrier technique findings between this study and Burger *et al*,[14] we may infer that ball-carrier technical deficiencies at youth level carry over to senior level. Although this reasoning highlights a potential training intervention point, a longitudinal study design, measuring ball-carrier's technical

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.

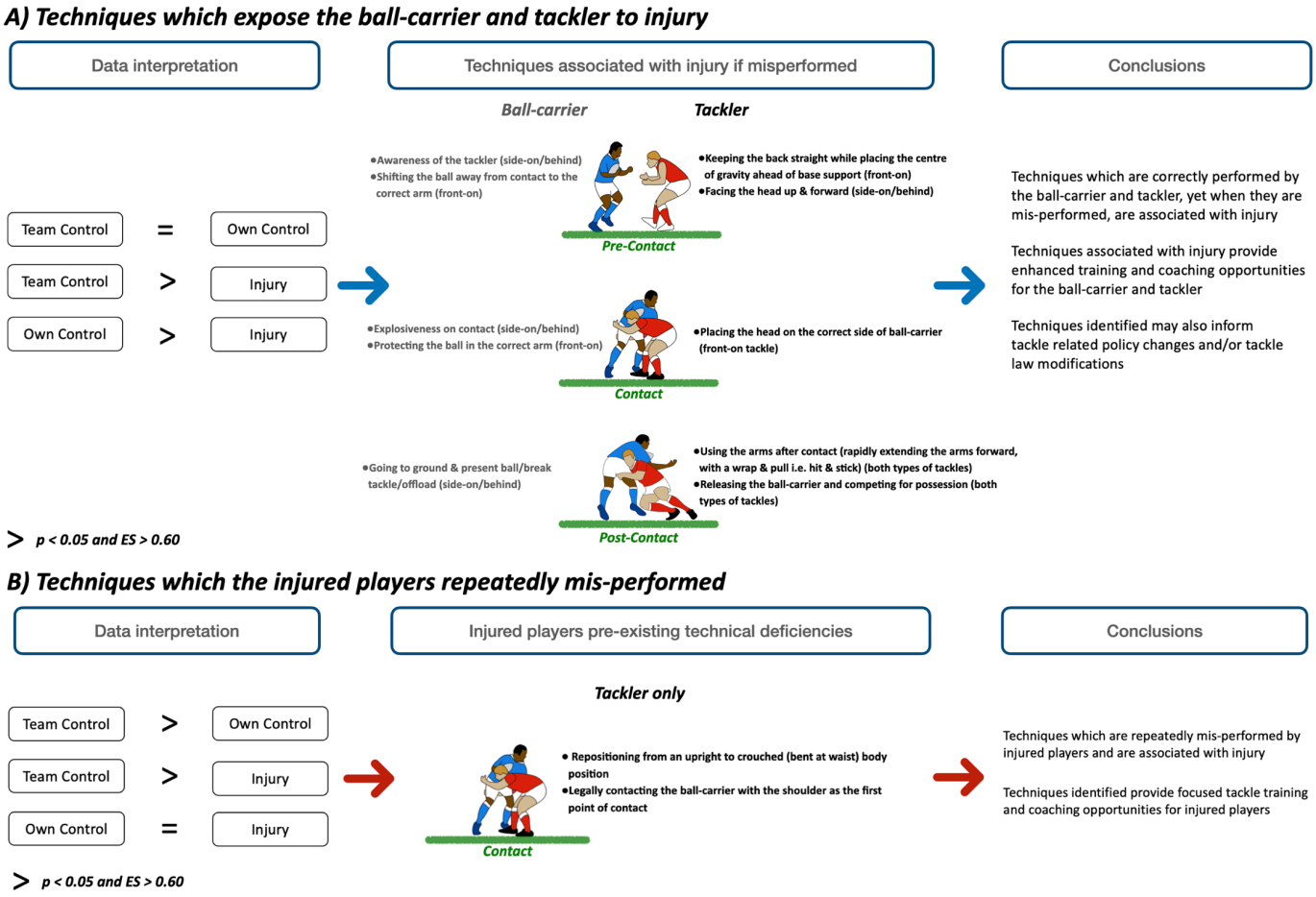

Figure 3 (A) Techniques which expose the ball-carrier and tackler to injury. (B) Techniques which the injured players repeatedly misperformed. ES, effect size.

proficiency over time (or across competitions), is required to prove/disapprove such a hypothesis.

Techniques which the injured players repeatedly misperformed

The purpose of including a team control was to help identify whether injured players had an existing technical deficiency which placed the player at risk of injury. To do this, we first compared the technical proficiency scores of each technique between the two non-injury control groups. Then we compared each of these control groups with the injured group. If there were moderate to large differences between the two control groups, and the team control and injured group, but trivial differences between the player's own control and injured group, we can infer that the injured players were underperforming in those specific techniques. Two front-on tackler techniques were identified using this process—repositioning from an upright to a crouched (bent at the waist) body position when making the tackle, and legally contacting the ball-carrier with the shoulder as the first point of contact. The inability of tacklers to perform these techniques support previous tackle-related head injury studies in elite rugby union, where upright tacklers have shown a propensity for injury.[23 36 37] For example, in a sample of 464 head injuries that occurred between 2013 and 2015 in six major professional rugby union competitions, Tucker *et al*[36] reported that an upright tackler was 1.5 times more likely to experience a head injury than a tackler bent at the waist. Also, a study examining the influence of tackle height on tackler head injury suggests tacklers should target the ball-carriers mid-trunk (base of chest/pectorals to top of pelvis) and lower trunk (top of pelvis to base of pelvis) to reduce their risk of injury.[38] In other words, to contact the mid-trunk or lower trunk of the ball-carrier, tacklers cannot be in an upright position. Simulations studies have also shown that contacting the ball-carrier in these areas may lower the inertial head loading experienced by the tackler.[39] Given the injury risk associated with an upright tackler body position, World Rugby, the sport's governing body, has implemented stricter rulings around high-tackles, and trialled a law-change to lower the height of a legal tackle to below the nipple line.[40 41] Data from the current study for the tackler suggests that failing to use their shoulder as the first point of contact with the ball-carrier is associated with an increased probability of injury. This finding is supported by previous research,[14] and tackling technique guidelines in national injury prevention programmes,[30 42 43] which state tacklers should be aiming to contact the ball-carrier with their shoulder to complete an effective, safe tackle. However, studies have also suggested that active shoulder tackles place both the tackler and ball-carrier at greater risk of concussion,[36] while tacklers have also been shown to have greater propensity to sustain head/neck injuries when using shoulder tackles.[22] When interpreting the difference in injury risk for shoulder tackles between these studies and the current findings, it is worth noting the dissimilarities in th methods. In the current study, the manner in which the tackler made shoulder contact, that is, active or passive, was not classified. Also, we did not compare shoulder tackles with other tackle types. That said,

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.

the potential of shoulder tackles to be protective and detrimental signifies the importance and delicate nature of this technique. It is the moment of first contact between the ball-carrier and tackler, and a product of a sequence of movements that occurred in the preparatory phase of the tackle (precontact). It is also the start of the post-tackle phase. If the tackler adequately prepares for the ensuing shoulder contact, the risk of injury may be reduced and the tackle can be safely and effectively completed. If the tackler does not adequately prepare for the ensuing shoulder contact and contact the ball-carrier with the shoulder correctly, they expose themselves to a high risk of head or neck injury.

Improving player welfare and injury prevention strategies

Using the specific tackler and ball-carrier techniques identified in this study in conjunction with the technical skill training tackle framework, coaches and trainers can design and implement focused tackle training programmes and sessions.[32] This will ensure tackle technique training sessions are optimised for learning and are effectively transferred to matches. In brief, coaches should start by refining and stabilising the player's coordination and movement patterns that form the technique(s). This initial stage should be performed in a training setting that is highly structured and controlled (low representativeness). To further stabilise the player's coordination and movement patterns, coaches can ask players to complete the technique(s) while fatigued. Once players are technically proficient and able to maintain good technique while fatigued, coaches can work towards a more representative and challenging training setting, and ask players to perform the techniques in response to the demands of the situation. To further enhance player's technique adaptability, and build technical capacity and robustness, players should then perform these same techniques in situations that are highly representative of match conditions.

Also, the findings from this study directly inform coaching curricula, which can assist coaches and trainers in optimising their tackle technique periodisation and training.[32 33] Players and coaches are more likely to accept and adopt injury prevention strategies if they also have performance benefits. The ball-carrier and tackler techniques identified in this study have also shown to have tackle performance benefits (such as gain line success and territorial dominance). For example, in a sample of 122 front-on tackles, Tierney *et al*[44] showed that the precontact tackler techniques upright to low, straight back with centre of gravity ahead of base support and shortening the steps increased the likelihood of the tackler winning the tackle contest.

Player, coach and referee education is central to the success of national injury prevention programmes such as New Zealand's Rugby Smart and South Africa's BokSmart.[31 45–47] This education is driven by research similar to that of the present study. As such, the findings of this study can also be used to inform and improve injury prevention educational resources and content. Professional rugby teams operate under challenging time constraints due to travel and other commitments, with set training schedules. As such, coaches, trainers and medical staff need to use their time efficiently when working with players. Although this study analysed the injured players as a group, the same methods can be used to analyse the technique of individual players. Therefore, in a team setting, building a database of technical proficiency scores for the team, both in training and matches, and comparing each player's scores with the team's score, will help to identify whether the player has an existing technical deficiency. Knowing which techniques to work on will facilitate more focused contact training sessions, thereby optimising coaches, trainers and medical staff time with the player. In the event when a tackle injury does occur, knowing which techniques the players were lacking will also assist the return to contact process, and reduce the risk of re-injury.

Limitations

The tackle is a physically and technically demanding open skill. Ball-carrier and tackler technique may be influenced by situational factors such as a player's physical attributes, shoulder dominance, playing position, match context and the opposing player's actions.[48–50] For example, Davidow *et al*[18] recently found that during head impact tackles, there is a technical deficiency in both the ball-carrier and tackler, irrespective of which player was injured, suggesting that the injured player alone is not always at fault from a technique perspective. As such, scoring the perfect technique in matches is highly challenging. Also,

What are the findings?

- Techniques which exposed tacklers to moderate and severe tackler injuries when misperformed were: keeping the back straight (front-on) while placing the centre of gravity ahead of base support (front-on), placing the head on correct side of ball-carrier (front-on tackle), facing the head up and forward (side-on/behind), using the arms after contact (rapidly extending the arms forward, with a wrap and pull, ie, hit and stick) (both front-on and side-on/behind tackles) and releasing the ball-carrier and competing for possession (both front-on and side-on/behind tackles).
- Techniques which exposed ball-carriers to moderate and severe ball-carrier injuries when misperformed were: awareness of the tackler (side-on/behind), shifting the ball away from contact to the correct arm (front-on), explosiveness on contact (side-on/behind), protecting the ball in the correct arm (front-on) and going to ground and present ball/break tackle/offload (both front-on and side-on/behind tackles).
- Techniques which the injured players repeatedly misperformed were: repositioning from an upright to crouched/bent at the waist body position, and connecting the ball-carrier with the shoulder as the first point of contact followed by use of arms to wrap the ball-carrier.

How might it impact on clinical practice in the future?

- Building a database of technical proficiency scores for the tackler and ball-carrier for the team, both in training and matches, and comparing each player's scores with the team's score, will help in identifying whether the player has an existing technical deficiency.
- Knowing which techniques to work on will facilitate more focused tackler and ball-carrier training sessions, thereby optimising coaches, trainers and medical staff time with the player.
- Correcting and strengthening players technical deficiencies that lead to an increased risk of injury during contact events will reduce the team's injury count and increase player availability for matches.
- In the event when a tackle injury does occur, knowing which techniques the player was lacking will also assist the return to contact process, and reduce the risk of re-injury.

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.

considering the influence of these situational factors on technique, the univariate nature of our analysis can be considered a limitation. As such, future studies in this area should endeavour to capture this information and apply more sophisticated multivariate statistical analyses. While the scoring technique approach based on its ideal form has shown promising construct validity,[14 15 18 49] and the current study showed good reliability, a level of subjectivity, which is inherent when scoring technique, should be acknowledged.

Although not a limitation, but worth noting for video analyses studies using injury surveillance data, is the challenge of identifying the injury events. Of the 288 moderate and severe contact injuries, 108 were identified in the video—87 of these were tackle related, and 74 met our criteria for technical proficiency analyses. Reasons we could not identify more injury cases were either one or a combination of the following—the injury may have occurred outside the view of the camera, poor camera angle or players not involved in the tackle occluding the view of the injury event. Also, in some cases, no injury event was visible in the video from what was recorded in the injury data. This may be because the player continued play and reported the injury after the match. Recently, the Rugby Union Video Analysis Consensus group recommended better integration between injury surveillance data and video, and the challenges outlined above can be used to argue for better integration—for example, having additional cameras focus on injury events only, and recording the approximate time the injury occurred (as opposed to match quarter only).[12]

CONCLUSION

This is the first study to analyse technique proficiency in professional rugby union matches for moderate and severe tackler and ball-carrier injuries. For the tackler and ball-carrier, for both front-on and side-on/behind tackles, overall technical proficiency scores in the injury-causing events were significantly lower when compared with the player's own injury-free tackles and the injury-free tackles of players from the same team. Using two controls we also identified which techniques injured players may be lacking, which puts them at risk of injury during a match. Injured tacklers were deficient in two front-on tackling techniques—repositioning from an upright to crouched/bent at the waist body position and connecting the ball-carrier with the shoulder as the first point of contact followed by use of arms to wrap the ball-carrier. Knowing which techniques to focus on during training, will improve how we train the tackle and optimise injury prevention strategies.

Twitter Steve den Hollander @steve_dh1, Michael Ian Lambert @MikeLambert01 and Sharief Hendricks @Sharief_H

Acknowledgements The authors would like to acknowledge all The Currie Cup Team Doctors, and Willie Maree and the South African Rugby Union Technical Department for providing the video footage. The authors would also like to thank Natalie Erskine @sportscicomm for the tackle illustrations.

Contributors VM and SH were responsible for drafting the manuscript. All authors contributed to data collection and analysis, and contributed to the final write-up of the manuscript.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not required.

Ethics approval The study was approved by the Human Research Ethics Committee of the University of Cape Town (HREC 437/2020).

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available on reasonable request.

ORCID iDs
Steve den Hollander http://orcid.org/0000-0002-6064-038X
Michael Ian Lambert http://orcid.org/0000-0001-8979-1504
Wayne Viljoen http://orcid.org/0000-0002-9432-6153
Sharief Hendricks http://orcid.org/0000-0002-3416-6266

REFERENCES

1 Fuller CW, Brooks JHM, Cancea RJ, *et al*. Contact events in rugby Union and their propensity to cause injury. *Br J Sports Med* 2007;41:862–7.
2 Roberts SP, Trewartha G, England M, *et al*. Collapsed scrums and collision tackles: what is the injury risk? *Br J Sports Med* 2015;49:536–40.
3 Williams S, Trewartha G, Kemp S, *et al*. A meta-analysis of injuries in senior men's professional rugby Union. *Sports Med* 2013;43:1043–55.
4 Brooks JHM, Fuller CW, Kemp SPT, *et al*. Epidemiology of injuries in English professional rugby Union: Part 1 match injuries. *Br J Sports Med* 2005;39:757–66.
5 West SW, Starling L, Kemp S, *et al*. Trends in match injury risk in professional male rugby Union: a 16-season review of 10 851 match injuries in the English Premiership (2002-2019): the professional rugby injury surveillance project. *Br J Sports Med* 2021;55:676–82.
6 Eliakim E, Morgulev E, Lidor R, *et al*. Estimation of injury costs: financial damage of English premier League teams' underachievement due to injuries. *BMJ Open Sport Exerc Med* 2020;6:e000675.
7 Ardern CL, Österberg A, Tagesson S, *et al*. The impact of psychological readiness to return to sport and recreational activities after anterior cruciate ligament reconstruction. *Br J Sports Med* 2014;48:1613–9.
8 Williams S, Trewartha G, Kemp SPT, *et al*. Time loss injuries compromise team success in elite rugby Union: a 7-year prospective study. *Br J Sports Med* 2016;50:651–6.
9 Fuller CW. Implications of health and safety legislation for the professional sportsperson. *Br J Sports Med* 1995;29:5–9.
10 Bahr R, Krosshaug T. Understanding injury mechanisms: a key component of preventing injuries in sport. *Br J Sports Med* 2005;39:324–9.
11 Krosshaug T, Andersen TE, Olsen O-EO, *et al*. Research approaches to describe the mechanisms of injuries in sport: limitations and possibilities. *Br J Sports Med* 2005;39:330–9.
12 Hendricks S, Till K, den Hollander S, *et al*. Consensus on a video analysis framework of descriptors and definitions by the rugby Union video analysis consensus group. *Br J Sports Med* 2020;54:566–72.
13 Knudson DV. *Qualitative diagnosis of human movement: improving performance in sport and exercise*. Champaign: Human Kinetics, 2013.
14 Burger N, Lambert MI, Viljoen W, *et al*. Tackle technique and tackle-related injuries in high-level South African rugby Union under-18 players: real-match video analysis. *Br J Sports Med* 2016;50:932–8.
15 Hendricks S, O'connor S, Lambert M, *et al*. Contact technique and concussions in the South African under-18 Coca-Cola Craven week rugby tournament. *Eur J Sport Sci* 2015;15:557–64.
16 Tierney GJ, Denvir K, Farrell G, *et al*. The effect of tackler technique on head injury assessment risk in elite rugby Union. *Med Sci Sports Exerc* 2018;50:603–8.
17 Tierney GJ, Denvir K, Farrell G, *et al*. Does ball carrier technique influence tackler head injury assessment risk in elite rugby union? *J Sports Sci* 2019;37:262–7.
18 Davidow D, Quarrie K, Viljoen W, *et al*. Tackle technique of rugby union players during head impact tackles compared to injury free tackles. *J Sci Med Sport* 2018;21:1025–31.
19 Readhead C, Lambert M, Starling L. The Currie cup Premiership competition injury surveillance report 2019. *S Afri J Sports Med* 2020;32.
20 Fuller CW, Molloy MG, Bagate C, *et al*. Consensus statement on injury definitions and data collection procedures for studies of injuries in rugby Union. *Br J Sports Med* 2007;41:328–31.
21 Lewallen S, Courtright P. Epidemiology in practice: case-control studies. *Community Eye Health* 1998;11:57–8.
22 Fuller CW, Ashton T, Brooks JHM, *et al*. Injury risks associated with tackling in rugby Union. *Br J Sports Med* 2010;44:159–67.
23 Quarrie KL, Hopkins WG. Tackle injuries in professional rugby Union. *Am J Sports Med* 2008;36:1705–16.
24 den Hollander S, Lambert M, Jones B, *et al*. Tackle and ruck technique proficiency within Academy and senior Club rugby Union. *J Sports Sci* 2019;37:2578–87.
25 Hendricks S, Matthews B, Roode B, *et al*. Tackler characteristics associated with tackle performance in rugby Union. *Eur J Sport Sci* 2014;14:753–62.
26 Wheeler KW, Askew CD, Sayers MG. Effective attacking strategies in rugby Union. *Eur J Sport Sci* 2010;10:237–42.
27 Weir JP. Quantifying test-retest reliability using the intraclass correlation coefficient and the SEM. *J Strength Cond Res* 2005;19:231–40.
28 Hopkins WG. Measures of reliability in sports medicine and science. *Sports Med* 2000;30:1–15.

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.

29 Cohen J. *Statistical power analysis for the behavioral sciences*. 2nd ed. Hillsdale, NJ: Erlbaum, 1988.
30 Hendricks S, Lambert M. Tackling in rugby: coaching strategies for effective technique and injury prevention. *Int J Sports Sci Coach* 2010;5:117–35.
31 Burger N, Lambert M, Hendricks S. Lay of the land: narrative synthesis of tackle research in rugby Union and rugby sevens. *BMJ Open Sport Exerc Med* 2020;6:e000645.
32 Hendricks S, Till K, Oliver JL, *et al*. Technical skill training framework and skill load measurements for the rugby Union tackle. *Strength Cond J* 2018;40:44–59.
33 Hendricks S, Till K, Brown JC, *et al*. Rugby Union needs a contact skill-training programme. *Br J Sports Med* 2017;51:829–30.
34 West SW, Williams S, Kemp SPT, *et al*. Patterns of training volume and injury risk in elite rugby Union: an analysis of 1.5 million hours of training exposure over eleven seasons. *J Sports Sci* 2020;38:238–47.
35 Campbell PG, Peake JM, Minett GM. The specificity of rugby Union training sessions in preparation for match demands. *Int J Sports Physiol Perform* 2018;13:496–503.
36 Tucker R, Raftery M, Kemp S, *et al*. Risk factors for head injury events in professional rugby Union: a video analysis of 464 head injury events to inform proposed injury prevention strategies. *Br J Sports Med* 2017;51:1152–7.
37 Cross MJ, Tucker R, Raftery M, *et al*. Tackling concussion in professional rugby Union: a case–control study of tackle-based risk factors and recommendations for primary prevention. *Br J Sports Med* 2019;53:1021–5.
38 Tierney GJ, Simms CK. Can tackle height influence head injury assessment risk in elite rugby union? *J Sci Med Sport* 2018;21:1210–4.
39 Tierney GJ, Richter C, Denvir K, *et al*. Could lowering the tackle height in rugby Union reduce ball carrier inertial head kinematics? *J Biomech* 2018;72:29–36.
40 Raftery M, Tucker R, Falvey Éanna Cian. Getting tough on concussion: how welfare-driven law change may improve player safety-a rugby Union experience. *Br J Sports Med* 2020. doi:10.1136/bjsports-2019-101885. [Epub ahead of print: 12 Aug 2020].
41 Stokes KA, Locke D, Roberts S, *et al*. Does reducing the height of the tackle through law change in elite men's rugby Union (the Championship, England) reduce the incidence of concussion? A controlled study in 126 games. *Br J Sports Med* 2021;55:220–5. doi:10.1136/bjsports-2019-101557
42 New Zealand Rugby Union. *Technique: The key factors in the tackle and taking the ball into contact. RugbySmart : Good technique is safe technique*, 2008: 12–14.
43 Rugby SA. BokSmart: Safe and effective techniques in rugby - practical guidelines, 2009. Available: https://www.springboks.rugby/media/j05dfdhv/safe-rugby-techniques-practical-guidelines.pdf
44 Tierney GJ, Denvir K, Farrell G, *et al*. The effect of technique on tackle gainline success outcomes in elite level rugby Union. *Int J Sports Sci Coach* 2018;13:16–25.
45 Quarrie KL, Gianotti SM, Hopkins WG, *et al*. Effect of nationwide injury prevention programme on serious spinal injuries in New Zealand rugby Union: ecological study. *BMJ* 2007;334:1150.
46 Brown JC, Hendricks S, Lambert MI, *et al*. BokSmart rugby safety education courses are associated with improvements in behavioural determinants in attending coaches and Referees: presurvey-postsurvey study. *Inj Prev* 2020. doi:10.1136/injuryprev-2020-043903. [Epub ahead of print: 24 Aug 2020].
47 Viljoen W, Patricios J. BokSmart – implementing a national rugby safety programme. *Br J Sports Med* 2012;46:692–3.
48 Davidow D, Redman M, Lambert M, *et al*. The effect of physical fatigue on tackling technique in rugby Union. *J Sci Med Sport* 2020;23:1105–10.
49 Tierney GJ, Denvir K, Farrell G, *et al*. Does player time-in-game affect tackle technique in elite level rugby union? *J Sci Med Sport* 2018;21:221–5.
50 Sewry N, Lambert M, Roode B, *et al*. The relationship between playing situation, defence and tackle technique in rugby Union. *Int J Sports Sci Coach* 2015;10:1115–28.

Br J Sports Med: first published as 10.1136/bjsports-2020-103759 on 13 July 2021. Downloaded from http://bjsm.bmj.com/ on July 17, 2021 at Nationwide Children's Hospital. Protected by copyright.